CT553H · RESEARCH EXPLAINED

Residue-Level Degron Mutation Effect Prediction and Analysis Tool

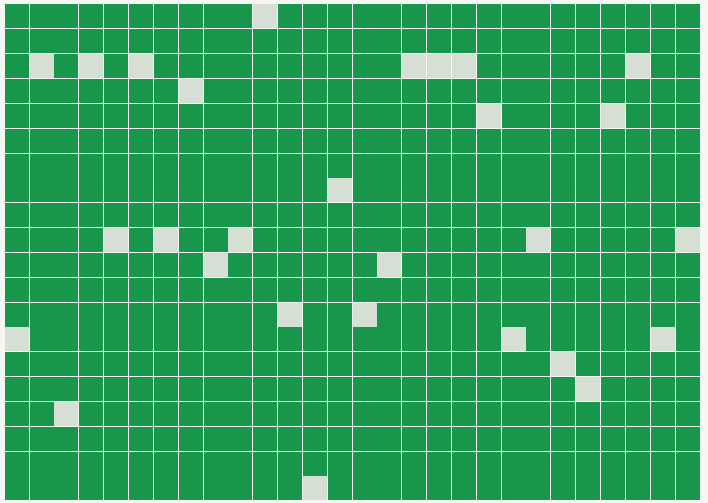
Machine learning to predict ΔPSI

A tool for prioritizing experiments

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Supervisor: Dr. Hồ Quang Thái

Can Tho University · 3 Nov 2026



One cell · one substitution

BIOLOGY AND THE PREDICTION TASK

A protein is built from an amino acid sequence



Protein structure model · illustration

Protein → peptide → residue

M	A	G	L	S	K	V	T

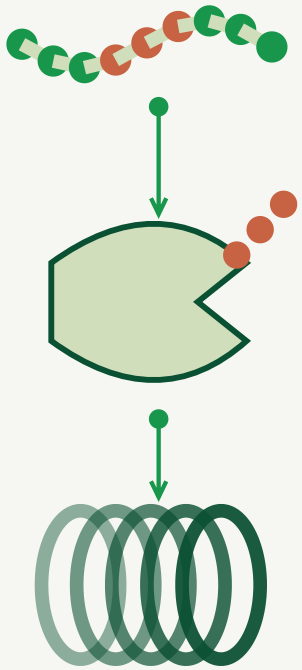
One letter · one amino acid at one position

A peptide is a short amino acid chain.
A residue is one amino acid unit in the chain.

The model reads a sequence of letters, each representing an amino acid.

BIOLOGY AND THE PREDICTION TASK

Degrons help direct protein degradation



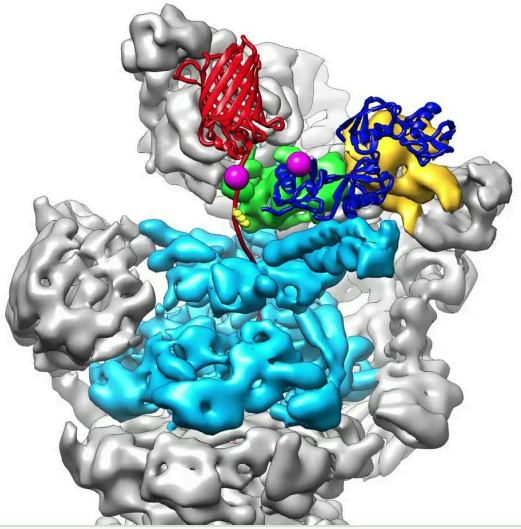
Protein with a degron

Recognition and ubiquitination

By the appropriate machinery

Proteasome

Cleavage into short peptides

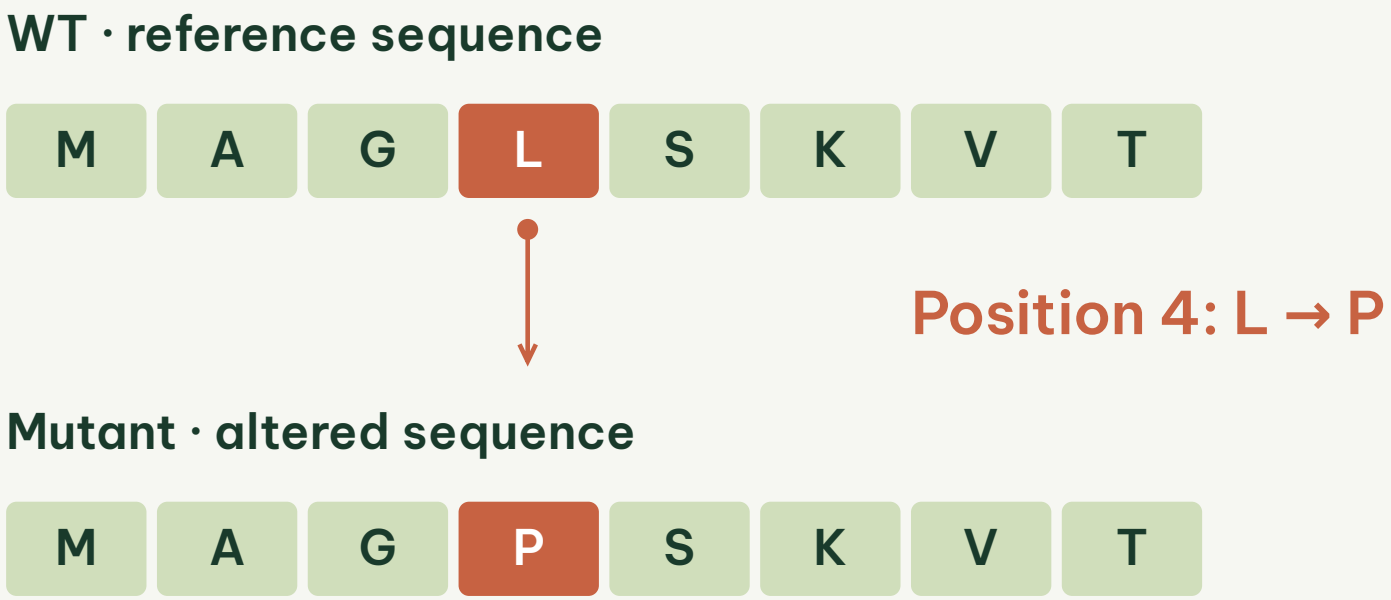


Andreas Martin / HHMI · NIGMS · molecular simulation

A degron is a signal that helps the appropriate machinery recognize a degradation substrate.

Recognition → ubiquitination → processing by the proteasome.

One substitution can change the signal



One substitution

position
reference AA
replacement AA

WT is the reference; the mutant differs at one amino acid.

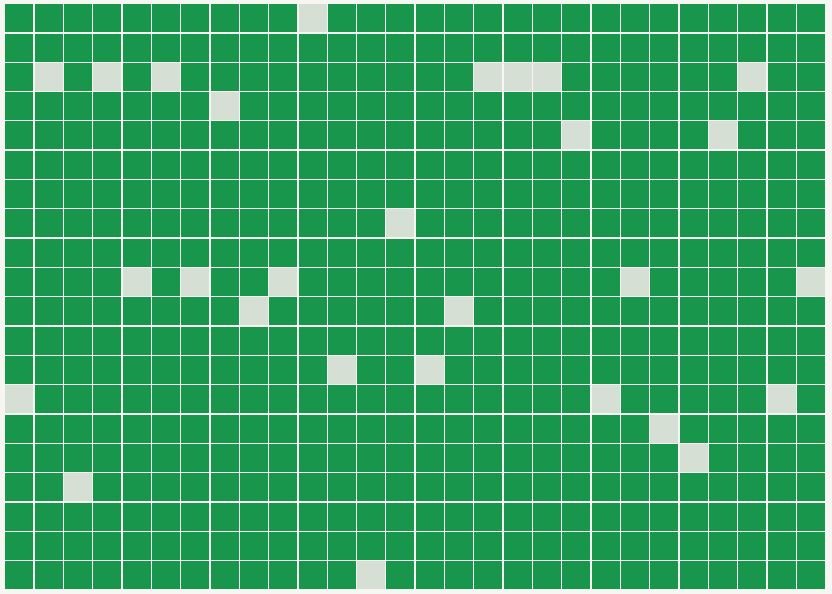
BIOLOGY AND THE PREDICTION TASK

532 substitutions motivate experimental prioritization

28 positions × 19 non-WT AAs

532

possible single substitutions



Each cell is one substitution

Segment activity alone does not identify which substitution to test first.

BIOLOGY AND EXPERIMENTAL LABELS

GPS converts fluorescence signals into PSI

Assay schematic · the histogram is illustrative

A shared reporter system



Reference

Signal reflects reporter stability

GFP / DsRed ratio



Six bins · from lower to higher stability

Normalized read distribution → PSI ∈ [1; 6]

Higher PSI means a more stable reporter in the Zhang assay.

BIOLOGY AND EXPERIMENTAL LABELS

ΔPSI is the effect the model aims to predict

Hypothetical example

Mutant **3.8** − WT **2.5**

ΔPSI = 3.8 − 2.5 = +1.3

Positive: the mutant reporter is more stable in this assay

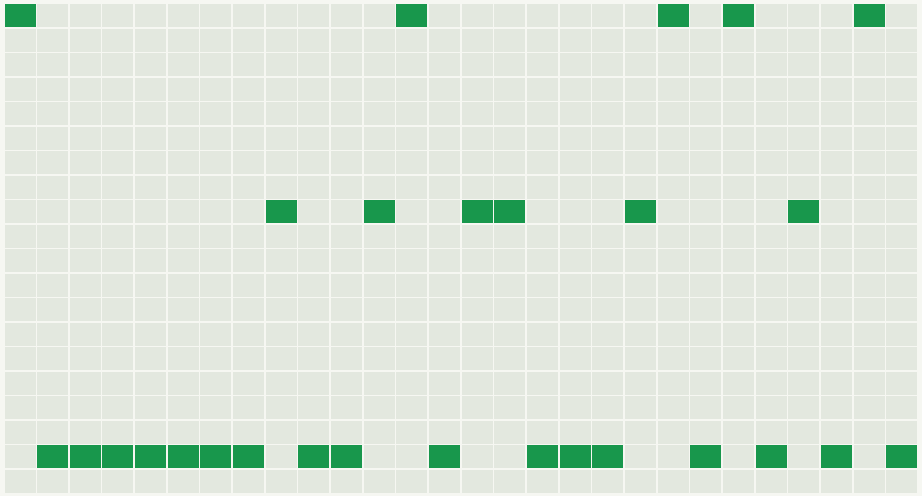
The output is a continuous value: a regression task.

DATA

Saturation and scanning provide complementary coverage

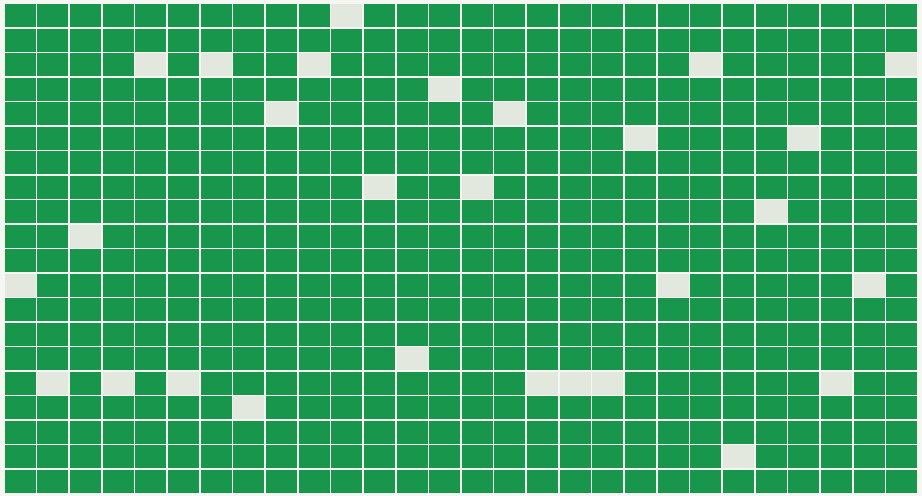
Design schematic · colored cells indicate assayed substitutions

Scanning



One substitution per position
20 WT → mutant types

Saturation



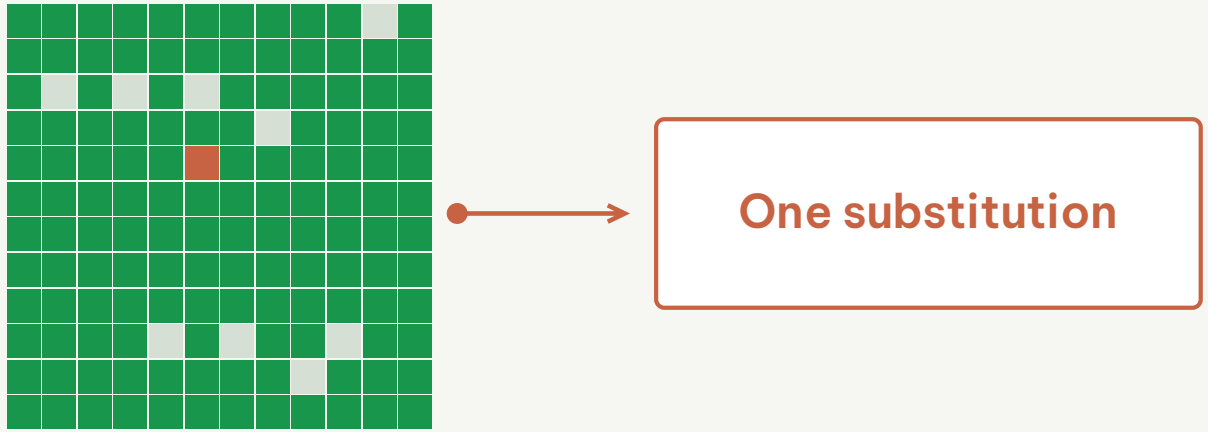
Many substitutions per position
380 WT → mutant types

Saturation covers diverse replacements; scanning spans more sequence contexts.

DATA

Each mutation cell becomes a labeled data record

Example record key · illustrative sequence values



Sample identity

Peptide	one sequence context
Position	4
Substitution	L → P
Label	PSI(mutant) − PSI(WT)

Match the peptide, position, substitution and WT reference before calculating ΔPSI.

DATA

The QC-filtered data retain the full PSI range

SATURATION

122,412

mutation cells · **235 peptides**

Benchmark for ΔPSI prediction

SCANNING

224,470

mutation cells after QC

Transfer learning and held-source evaluation

The 235 saturation peptides form **221 protein groups** in W.
Keep technical QC; use PSI thresholds to study training scope.

Mutation cells, peptides and protein groups are different counting units.

MACHINE LEARNING WORKFLOW

The pipeline connects data, models and the analysis tool

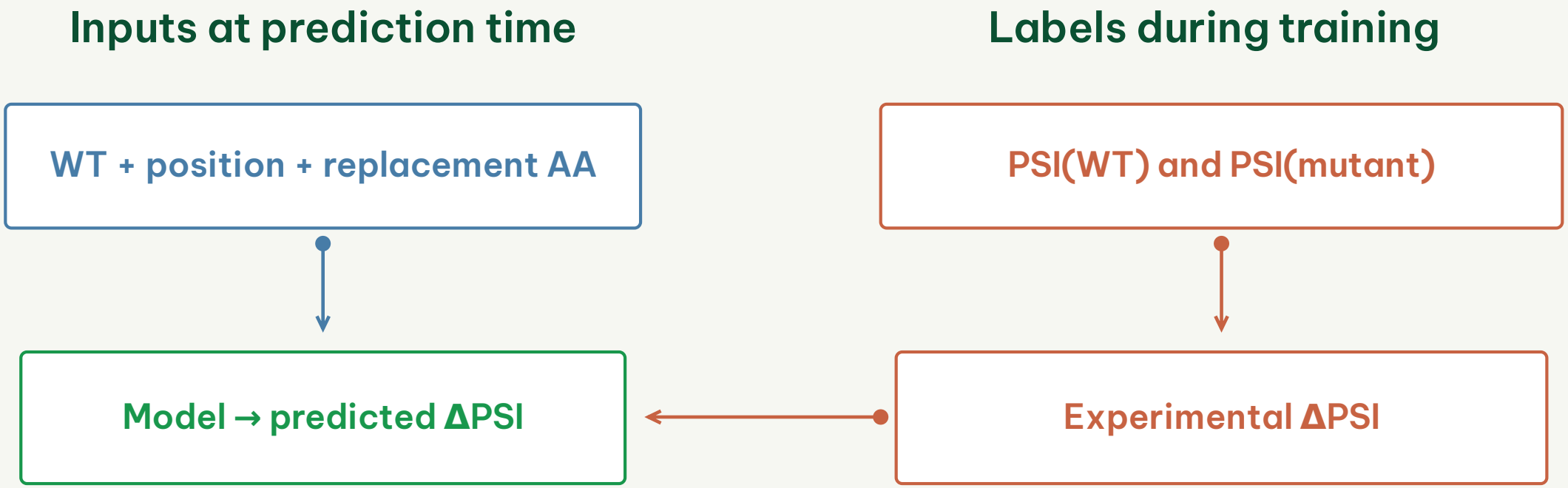


Group-based evaluation → Lock and validate → Analysis tool

Each processing step needs a rationale, an output and traceable evidence.

MACHINE LEARNING WORKFLOW

Sequences are inputs; PSI measurements define the labels

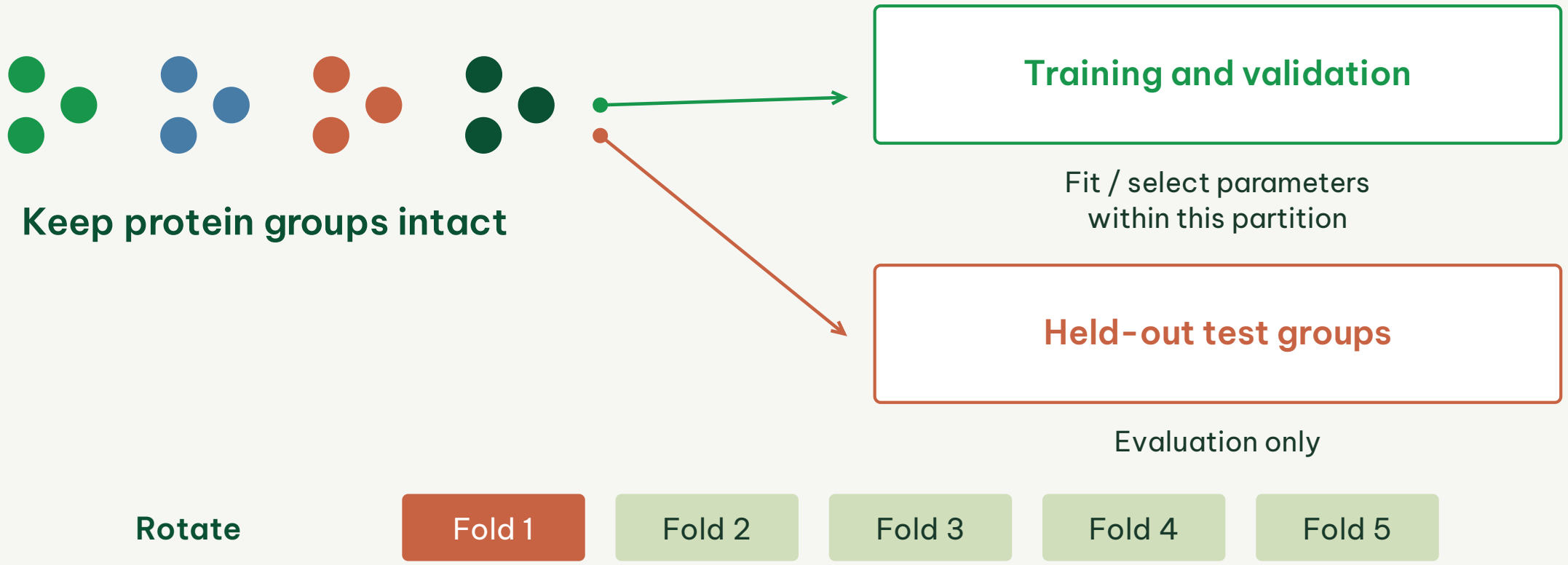


Labels guide learning; measured PSI is not an input feature

The model receives no experimental PSI when predicting a new substitution.

MACHINE LEARNING WORKFLOW

Keep protein groups intact when splitting the data



Five outer folds evaluate protein groups held out from training.

MODELS AND HYPOTHESES

Ridge tests the predictive value of sequence context



L → P

Substitution

WT and mutant one-hot

Composition

Frequencies of 20 AAs

Context ±5

Ordered nearby AAs

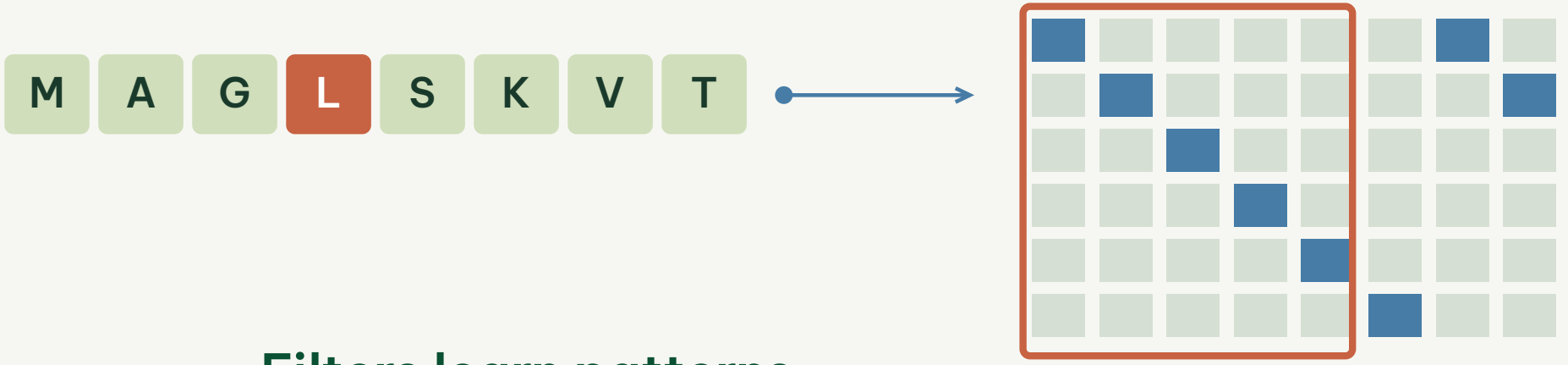
Features × weights → one ΔPSI value

Question: how much can sequence features and a linear model already predict?

MODELS AND HYPOTHESES

A one-hot CNN learns local sequence patterns

Simplified one-hot schematic · not the full numeric matrix



Filters learn patterns in local neighborhoods

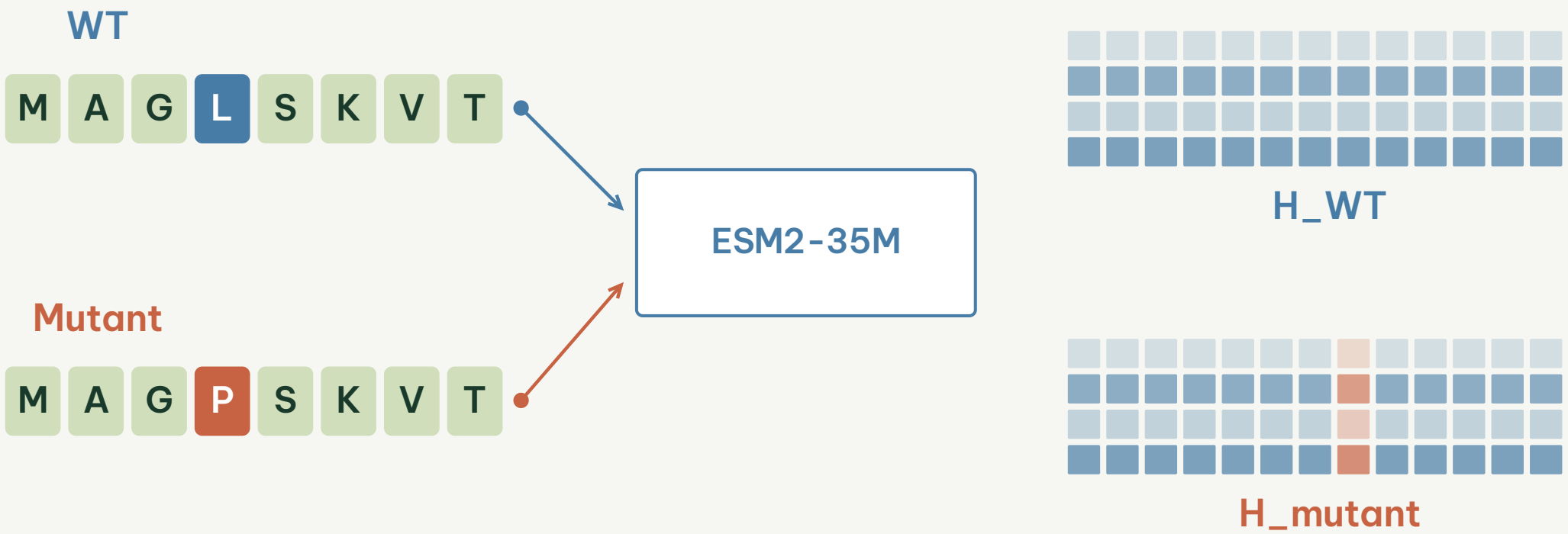
One-hot → CNN → features → ΔPSI

The one-hot control tests the added value of ESM representations.

MODELS AND HYPOTHESES

ESM2 converts sequences into numeric representations

Representation schematic · colored cells illustrate vectors

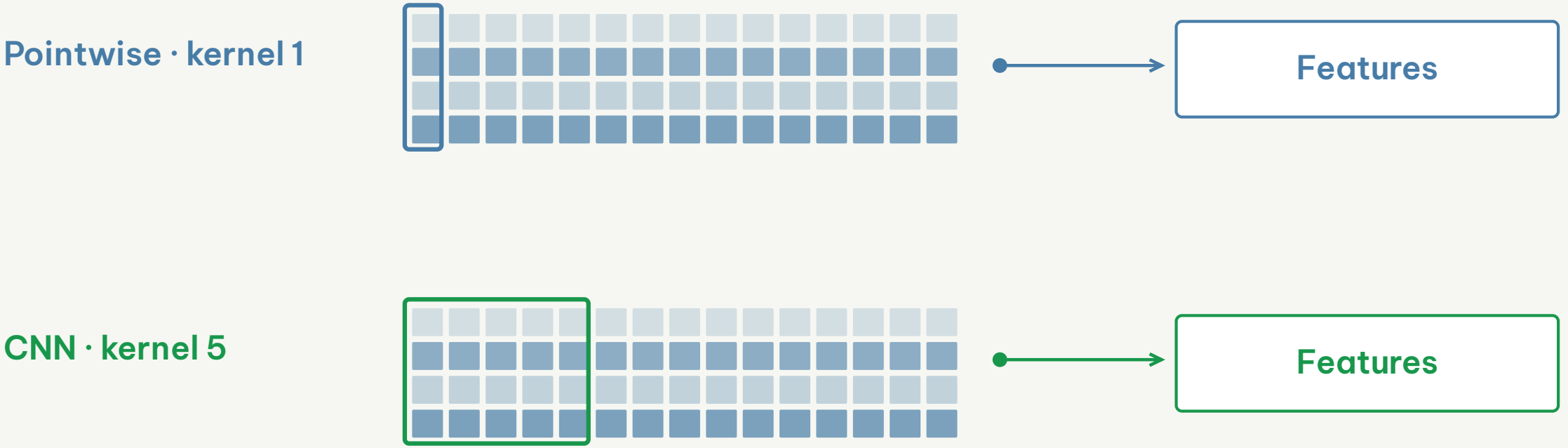


$$\Delta H = H_mutant - H_WT$$

Embedding differences describe representation changes between mutant and WT.

MODELS AND HYPOTHESES

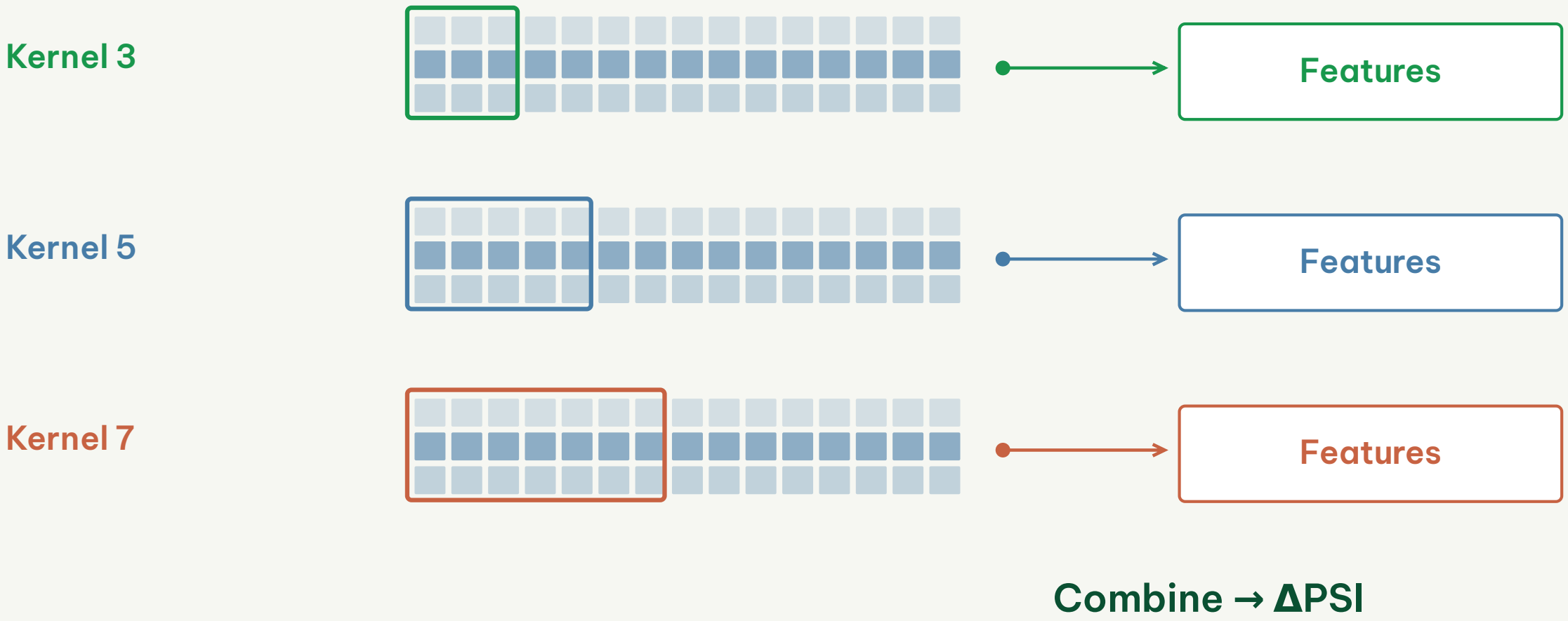
Pointwise and CNN heads differ in position mixing



ESM embeddings already contain context; kernel 5 adds position mixing in the head.

MODELS AND HYPOTHESES

mCNN explores multiple context widths



Question: do kernels 3, 5 and 7 improve prediction over a single kernel?

MODELS AND HYPOTHESES

Frozen and LoRA update different parameters



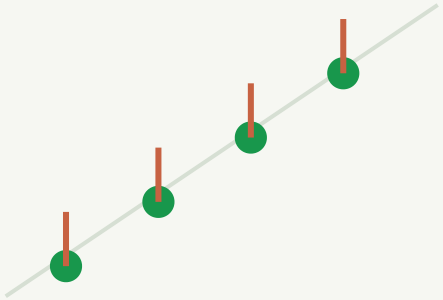
Match seeds, folds and protocol · assess error and cost

Assess PLM adaptation together with computational cost and stability.

EVALUATION

Error, ranking and confidence intervals answer different questions

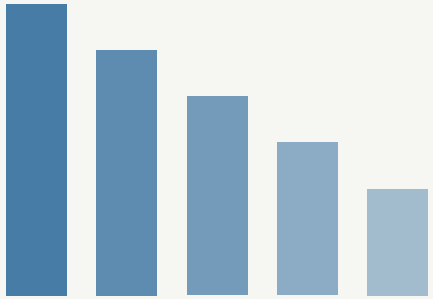
Error



MAE

Prediction–measurement error

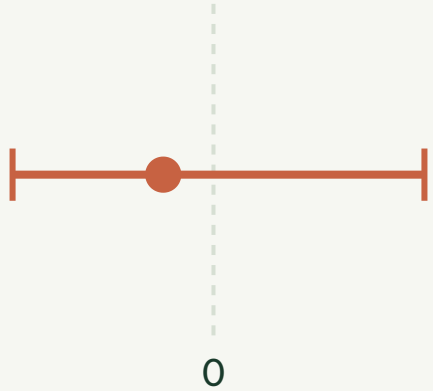
Ranking



AUROC / AP

Rank cells with ΔPSI ≥ 1

Uncertainty



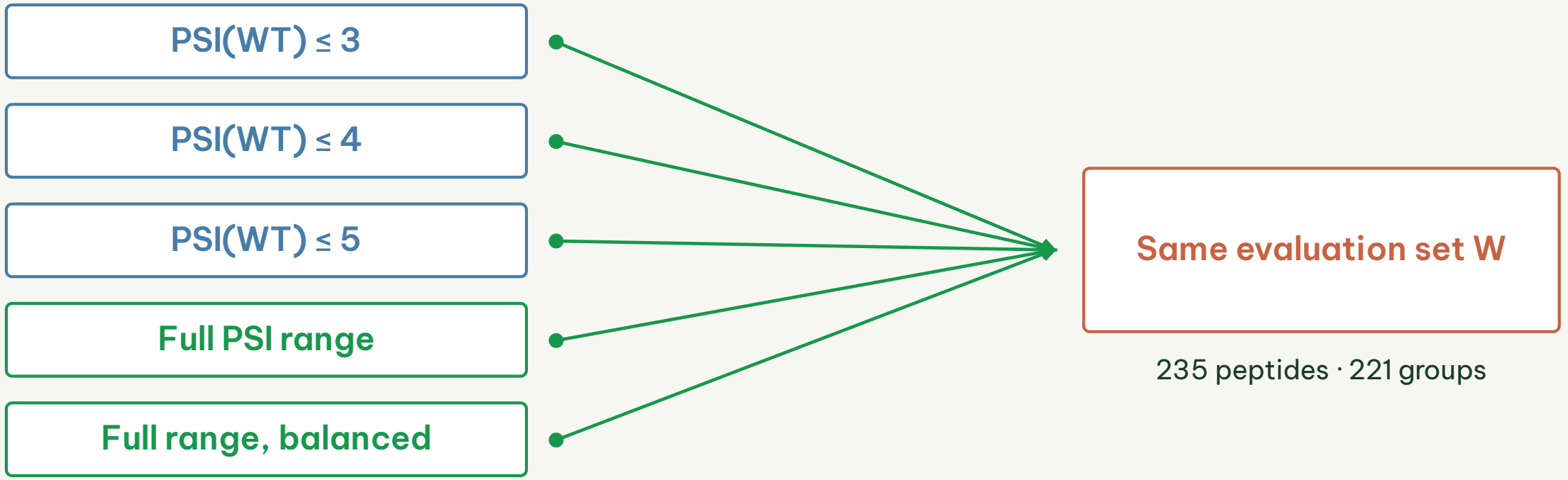
Confidence interval

Bootstrap by protein group

MAE measures regression error · AUROC/AP assess a supplementary positive event

Lower MAE is better; higher AUROC/AP is better on eligible cohorts.

Experiment A: which PSI range should be used for training?

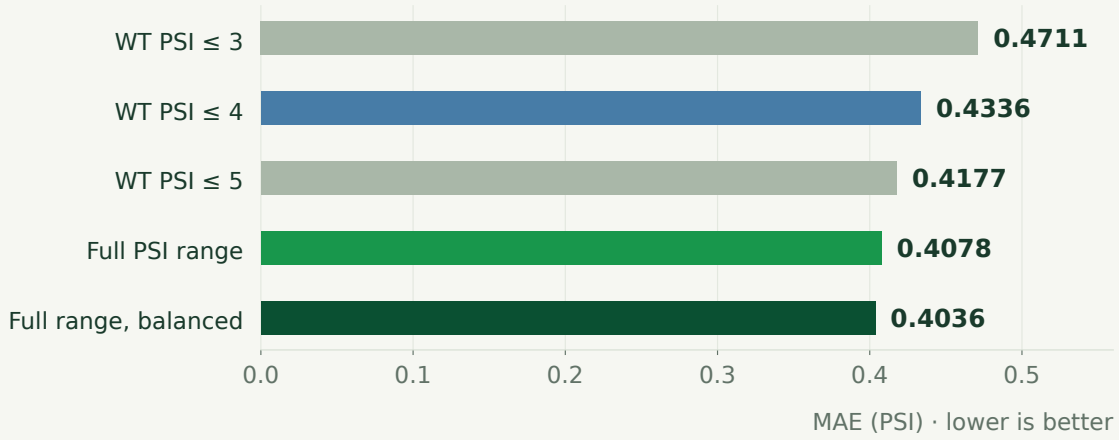


PSI thresholds define training scope; all configurations are compared on W.

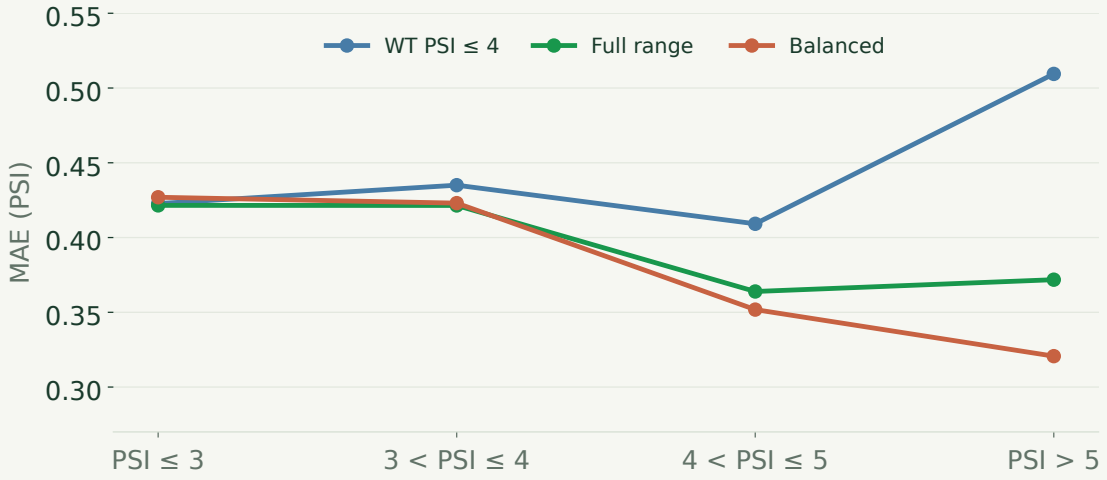
EXPERIMENTS AND RESULTS

Full-range training reduces Ridge regression error

Same evaluation set W · MAE ↓



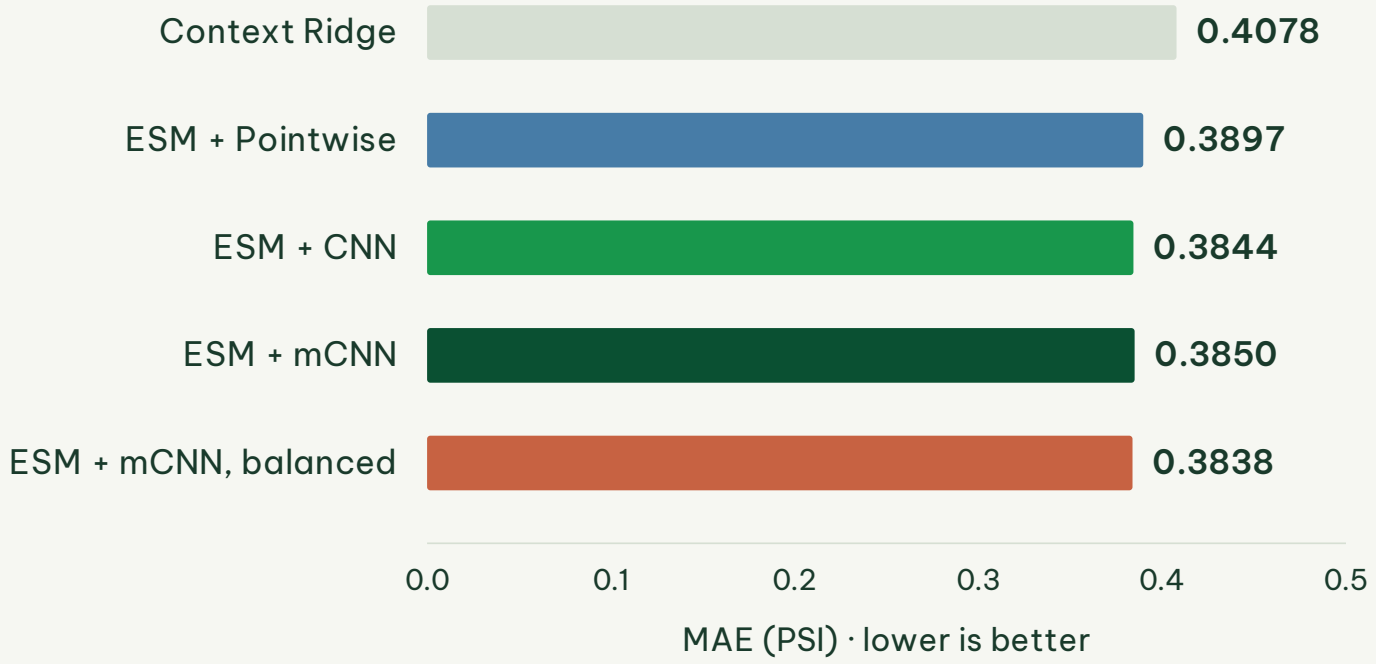
Benefits vary across test PSI strata



Full-range training reduces MAE by about **5.94%** relative to training only on PSI ≤ 4.

EXPERIMENTS AND RESULTS

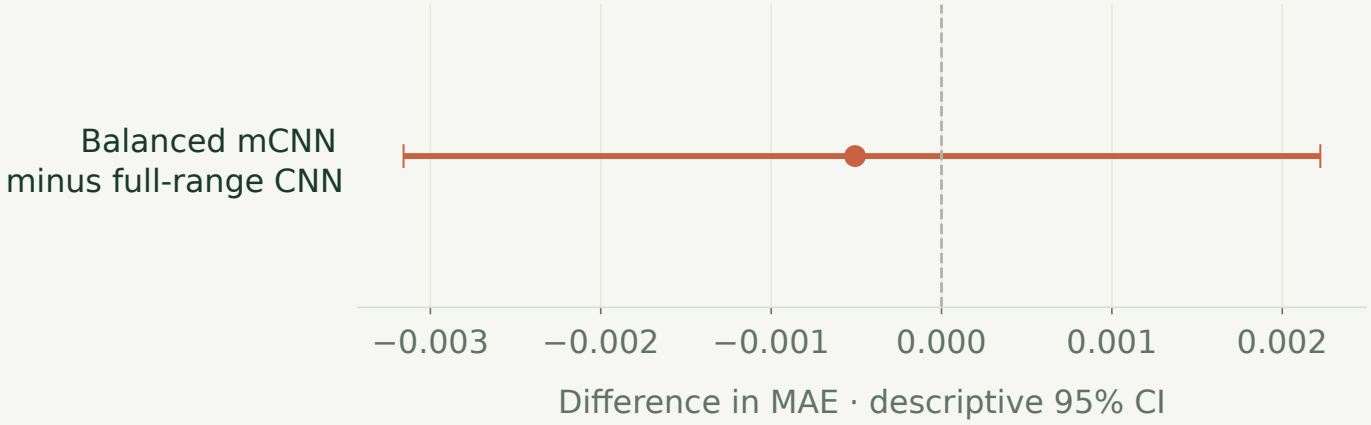
Experiment B: what do ESM2 and CNN add?



Same W · three-seed ensemble · ESM2-35M · development results.

EXPERIMENTS AND RESULTS

Small differences need confidence intervals



−0.000508

PSI · difference in MAE

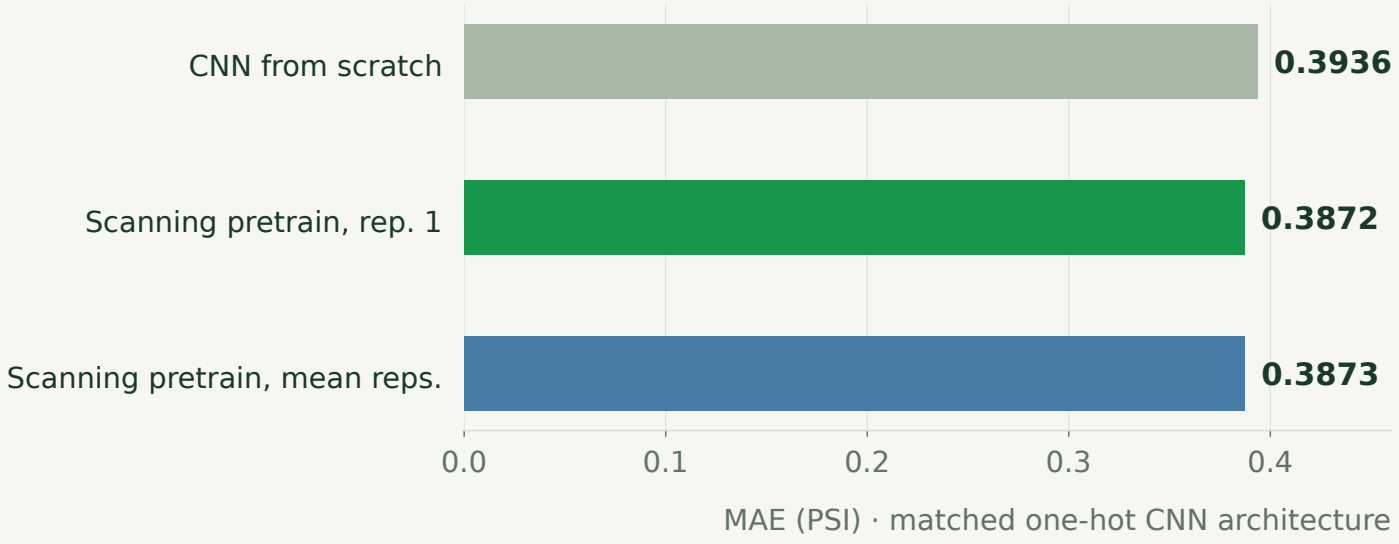
The CI crosses **0**.

Training weights also differ.

The evidence does not establish that mCNN outperforms CNN.

EXPERIMENTS AND RESULTS

Experiment C: open questions about transfer learning



Scanning → saturation
Improves on one-hot CNN training from scratch.

ESM + scanning
The combined approach was not evaluated in this snapshot.

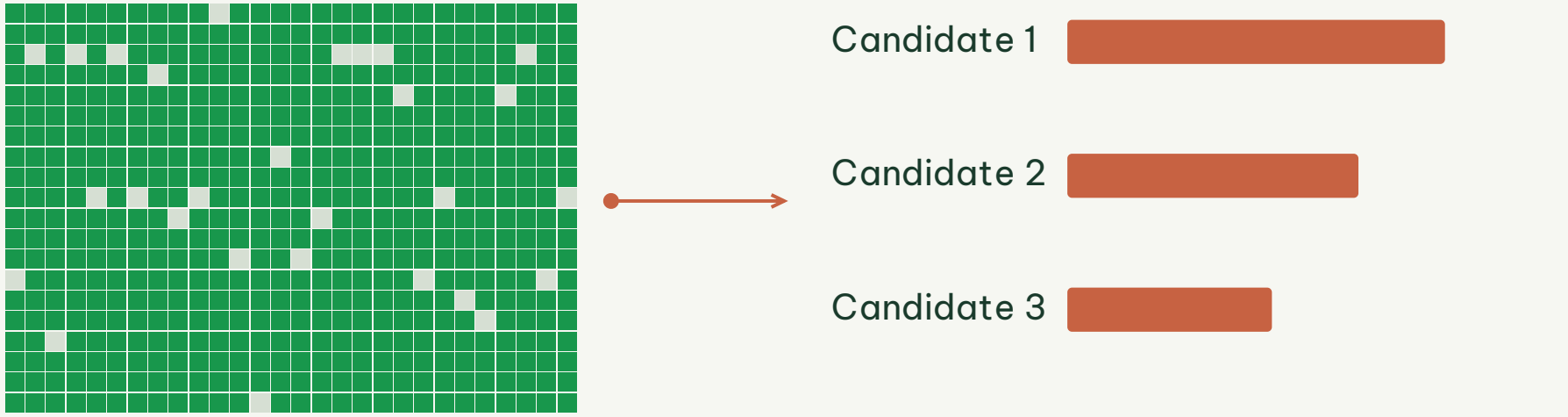
LoRA versus frozen
OOF results are incomplete.

Transfer benefits depend on inputs, design and matched controls.

ANALYSIS TOOL

The effect map supports experimental prioritization

Illustrative analysis workflow



Effect map

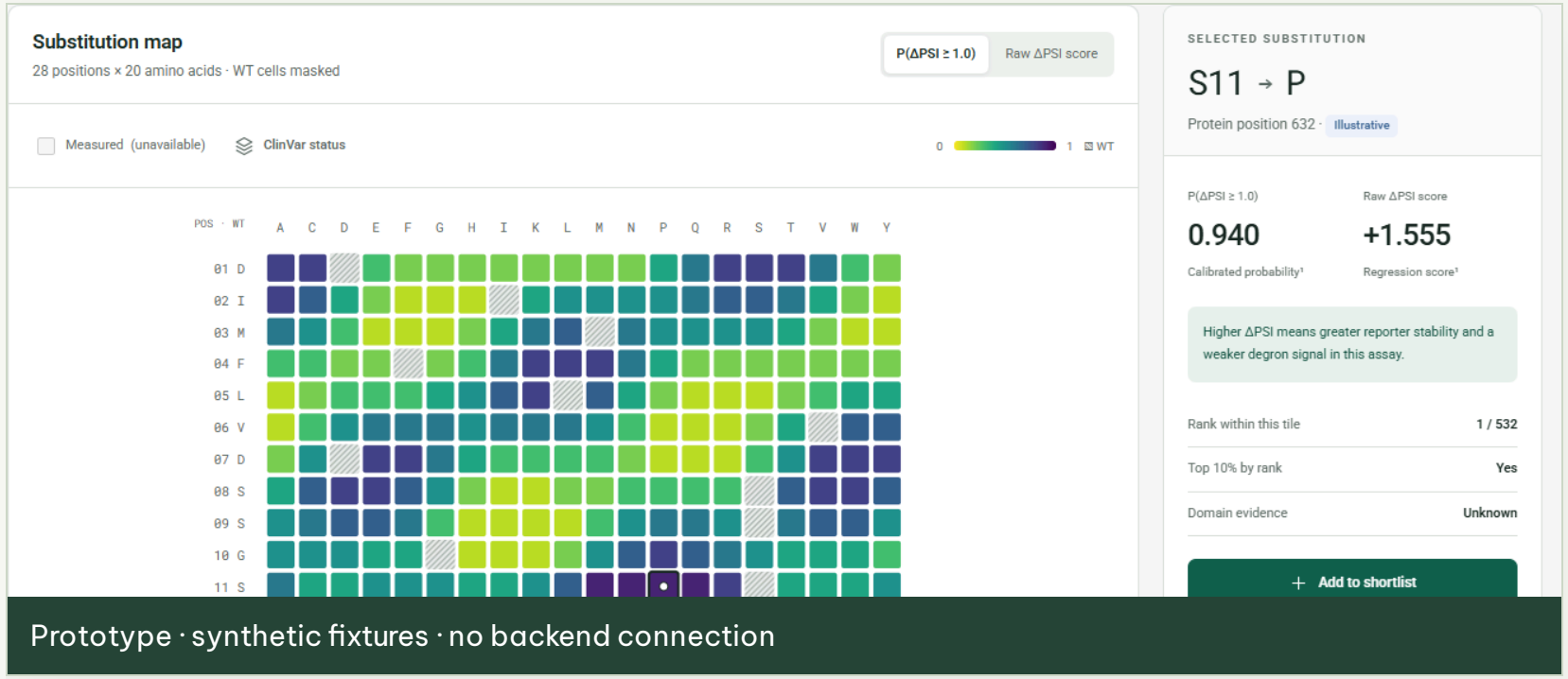
Select → validate experimentally

1 · Enter peptide | 2 · View map | 3 · Select variant | 4 · Inspect details | 5 · Export results

Predictions provide a shortlist for follow-up experiments.

ANALYSIS TOOL

The serving baseline and UI prototype have different roles



Prototype · synthetic fixtures · no backend connection

Implemented baseline

React + TypeScript

FastAPI

Serving model:

`m3-ext4.0-platt-v1`

New research models

Still require refitting, validation and integration.

Scores in the interface screenshot are synthetic fixtures.

STATUS AND CONTRIBUTIONS

What is available and what remains unfinished

Component	Documented status
Data, QC and group splits	Built, with traceability records
Ridge / ESM2 / CNN / mCNN	Development results available
PLM expansion / frozen–LoRA	Ongoing; no final conclusion
Calibration / T / external sources	Incomplete for the new candidate
FastAPI + React baseline	Implementation available
Workspace and new-model integration	Implementation / validation pending

Document snapshot: 3 Nov 2026 · research evidence and software progress are separate.

STATUS AND CONTRIBUTIONS

Contributions should be grounded in inspectable evidence

01

Data and benchmarks

Labels, QC, provenance and protein-group splits

02

Methodological evidence

PSI range, representations, heads and transfer learning

03

Research tool

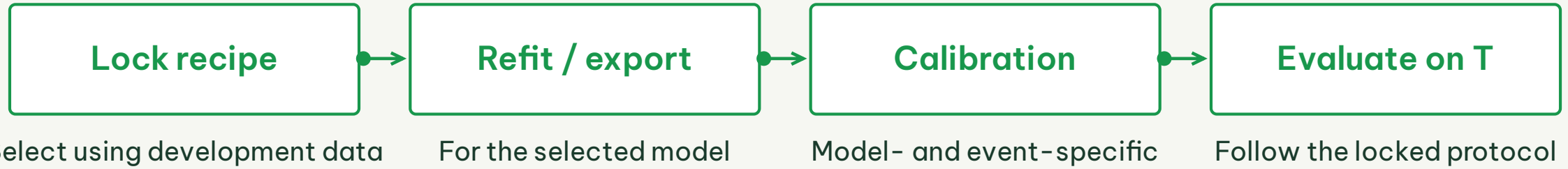
Effect maps, prioritization and traceable results

Research value requires empirical comparisons and clearly bounded conclusions.

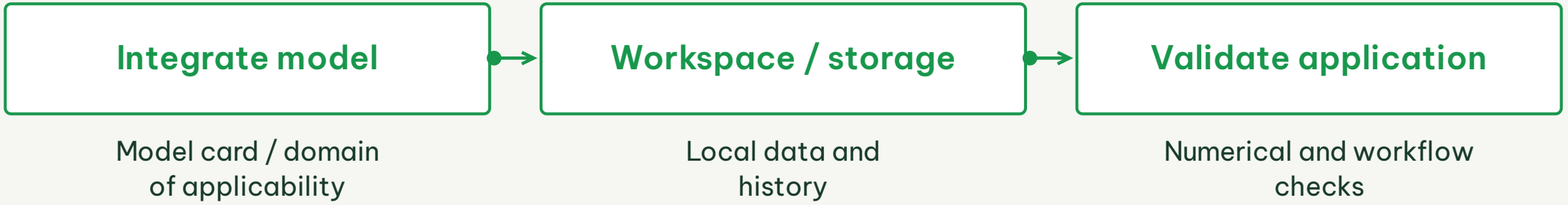
NEXT STEPS

Lock and validate the model before integration

MODEL RESEARCH



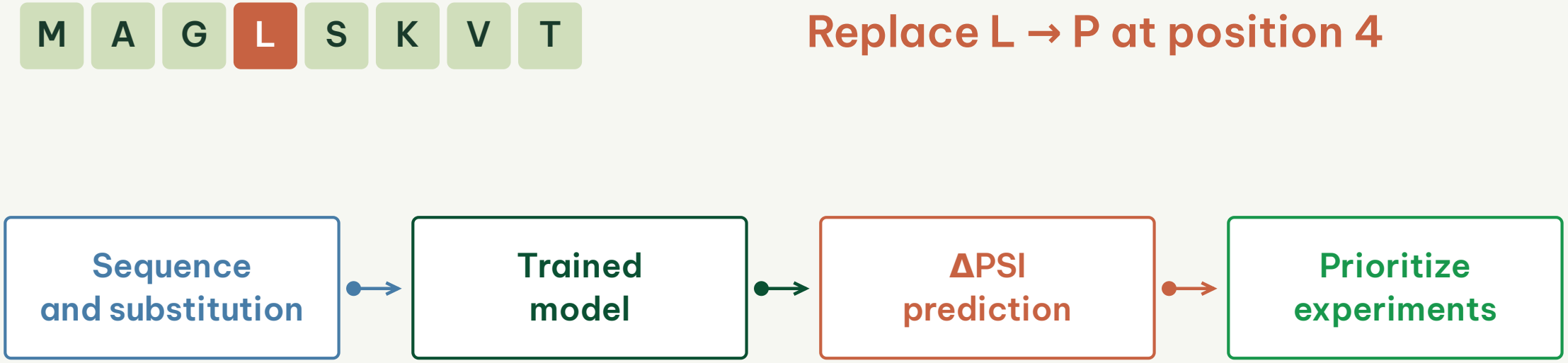
TOOL DEVELOPMENT



PLM expansion and LoRA need complete comparisons; new models need a serving workflow.

THE COMPLETE RESEARCH QUESTION

From a sequence to a testable hypothesis



Available evidence: a development benchmark on Zhang data

Aim: predict the effects of individual substitutions to support research.

APPENDIX · TERMINOLOGY

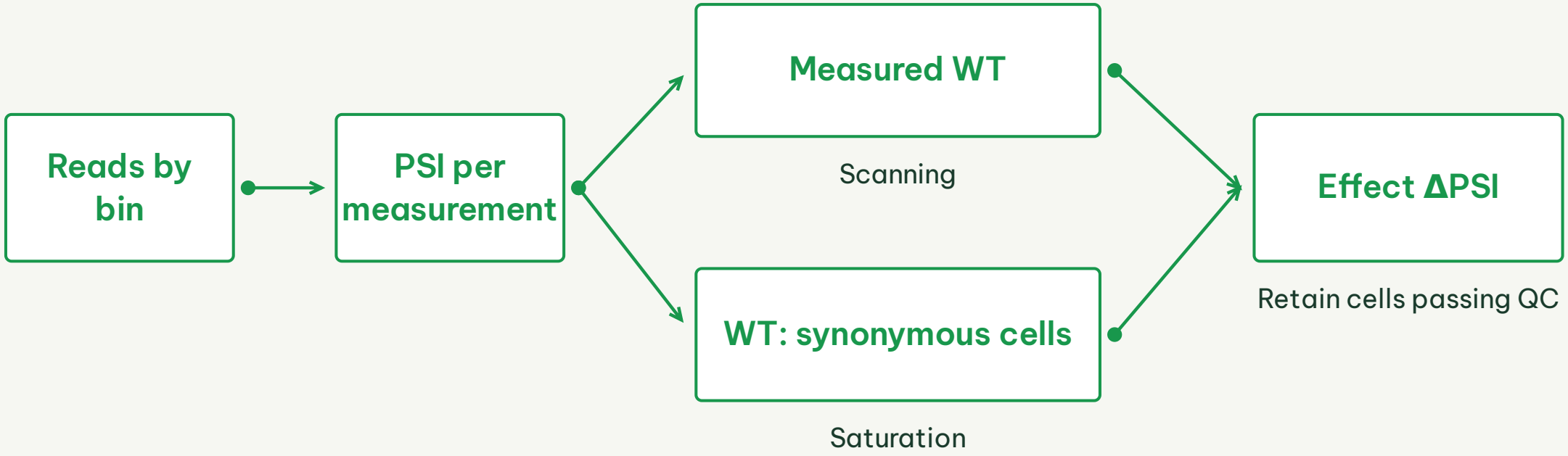
Key terms used in this presentation

Term	Meaning in this project
Residue / peptide	One amino acid unit in a chain / a short chain
WT / mutant	Reference sequence / sequence after substitution
PSI / ΔPSI	Reporter stability index / change relative to WT
Embedding / head	Numeric representation / the prediction module using it
Fold / OOF	A group partition / prediction with the sample held out
Calibration / CI	Probability calibration / statistical confidence interval

Reporter stability is the experimental endpoint used here.

APPENDIX · LABEL CONSTRUCTION

From read counts to ΔPSI: use the correct WT reference

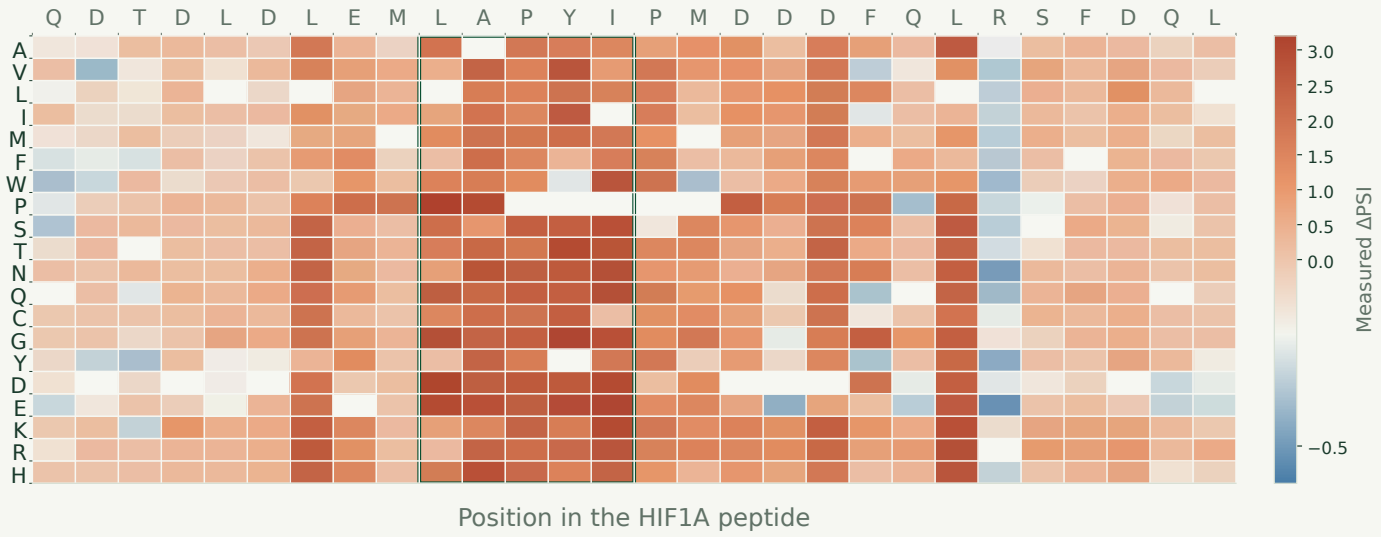


$$PSI = \Sigma(bin \times normalized\ reads) / \Sigma(normalized\ reads)$$

Check read depth, replicates, sequences and coordinates before retaining cells.

APPENDIX · EXPERIMENTAL EXAMPLE

HIF1A example: reading an experimental effect map



529 / 532

substitutions with measured labels

Columns: peptide positions
Rows: replacement AAs
Color: ΔPSI value

Experimental labels

Different positions can respond very differently to substitutions.

APPENDIX · METRICS AND CIS

Results and units of evaluation

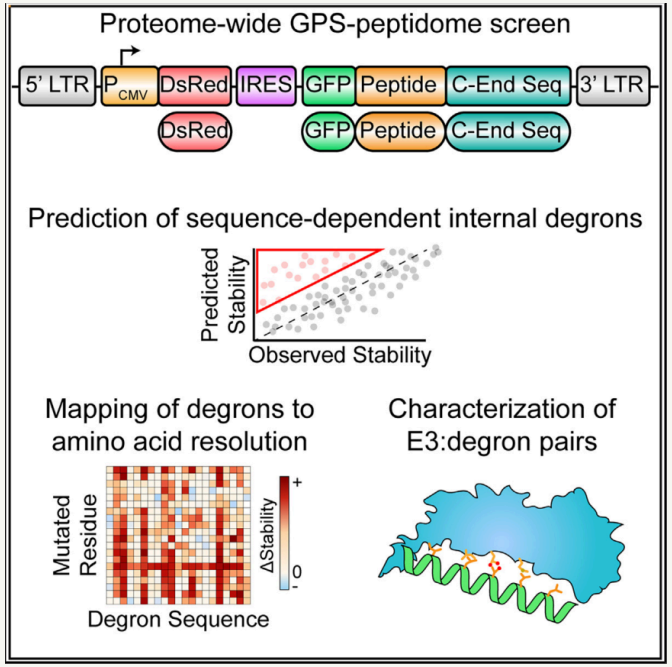
Model / training scope	MAE ↓	Positive-event AUROC ↑	Positive-event AP ↑
Context Ridge	0.4078	0.8319	0.4614
ESM + Pointwise	0.3897	0.8375	0.4739
ESM + CNN	0.3844	0.8488	0.5095
ESM + mCNN	0.3850	0.8463	0.5042
ESM + mCNN, balanced	0.3838	0.8470	0.5064

MAE: 235 peptides / 221 groups · positive-event AUROC and AP: 184 eligible peptides
Ensemble seeds 17, 29, 43 · five outer folds · protein-group bootstrap CIs

The first four rows use full-range training; balanced mCNN changes training weights.

APPENDIX · SOURCES

Sources and evidence can be explored by topic



Zhang et al., 2023 · Graphical abstract · CC BY 4.0

Original paper and data
Zhang et al. · Molecular Cell 2023

Data and QC-filtered counts
Mutation cells, peptides and protein groups

Results and comparison conditions
Kaggle snapshot · analysis dated 2 Nov

External sources and baseline models
PAP and Degformer audits; historical records

Software scope
Baseline, prototype and outstanding work

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Open “Notes” for the current slide's sources and limitations.